

Detecting the potential active pocket of protein by p2rank, fpocket, and MolAICal

Qifeng Bai

Email: molaical@yeah.net

Homepage: <https://molaical.github.io> or <https://molaical.gitlab.io>

School of Basic Medical Sciences

Lanzhou University

Lanzhou, Gansu 730000, P. R. China

1. Introduction

When a protein receptor has no bound ligand or no reported active pocket, it becomes necessary to predict its active site. P2Rank [1] and fpocket [2] are two popular open-source tools capable of identifying potential active pockets in receptors. fpocket is primarily based on geometric and topological principles (specifically, the Voronoi tessellation method), whereas P2Rank relies on machine learning, integrating local surface features and clustering techniques as its core methodology. fpocket is well-suited for rapid and large-scale screening of protein structure databases. In contrast, P2Rank is more appropriate when higher prediction accuracy, machine learning-based refinement, and prioritization of top candidate binding sites are desired.

2. Materials

2.1. Software requirement

1) MolAICal: <https://molaical.github.io>

2) PyMOL: <https://github.com/cgohlke/pymol-open-source-wheels>

The "pip" is required for the installation of PyMOL. Users can firstly install Miniconda (<https://www.anaconda.com/docs/getting-started/miniconda/install>), and then run the command below:

```
#> pip install pymol-3.2.0a0-cp314-cp314-win_amd64.whl
```

```
#> pip install pymol_launcher-3.0-cp314-cp314-win_amd64.whl
```

Finally, search for the keyword "pymol.exe". Once users find the "pymol.exe", create it as a desktop shortcut. Click it on the desktop, and PyMOL will open.

2.2. Example files

1) All the necessary tutorial files are downloaded from:

https://github.com/MolAICal/tutorials/tree/master/019-detect_pocket

3. Procedure

3.1. Detect the pocket by p2rank and MolAICal

Go to the folder [019-detect_pocket](#), and run the command below:

```
#> molaical.exe -call run -c pocket -i predict -f 1UYD.pdb -o results
```

The results will be in the folder "results". Firstly, input command "[cd 019-detect_pocket\results\visualizations](#)" in PyMOL and go to the target folder. And then, open the file "1UYD.pdb.pml" in the directory "results\visualizations" by PyMOL, the potential pockets will be shown (see Figure 1). It is evident that the top-ranked active pocket identified represents the ligand-binding region.

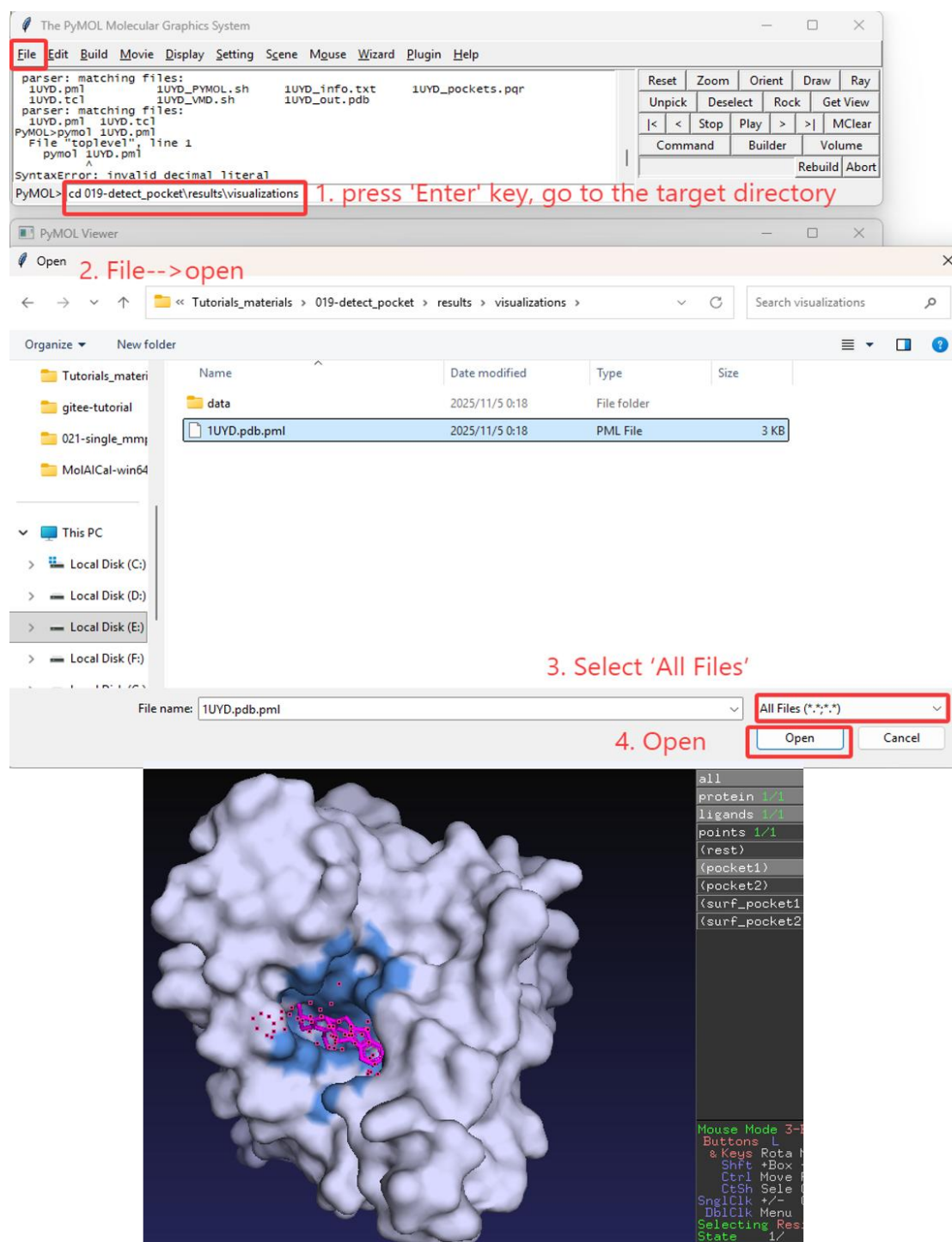


Figure 1

3.2. Detect the pocket by fpocket and MolaICal

Go to the folder [019-detect_pocket](#):

The fpocket is only run on Linux systems currently. Run the command below:

```
#> molaical.exe -call run -c fpocket -i -f 1UYD.pdb
```

The results will be in the folder "1UYD_out". In a similar way above, firstly, input command "`cd 019-detect_pocket\1UYD_out`" in PyMOL and go to the target folder. And then, open the file "1UYD.pml" in the directory "019-detect_pocket\1UYD_out" by PyMOL, the potential pockets will be shown (see Figure 2). And loading the ligand in the "1UYD.pdb" file as well. It shows that the top-rank pocket also contains the ligand-bound sites. It indicates that the fpocket also finds the right pocket of the protein receptor.

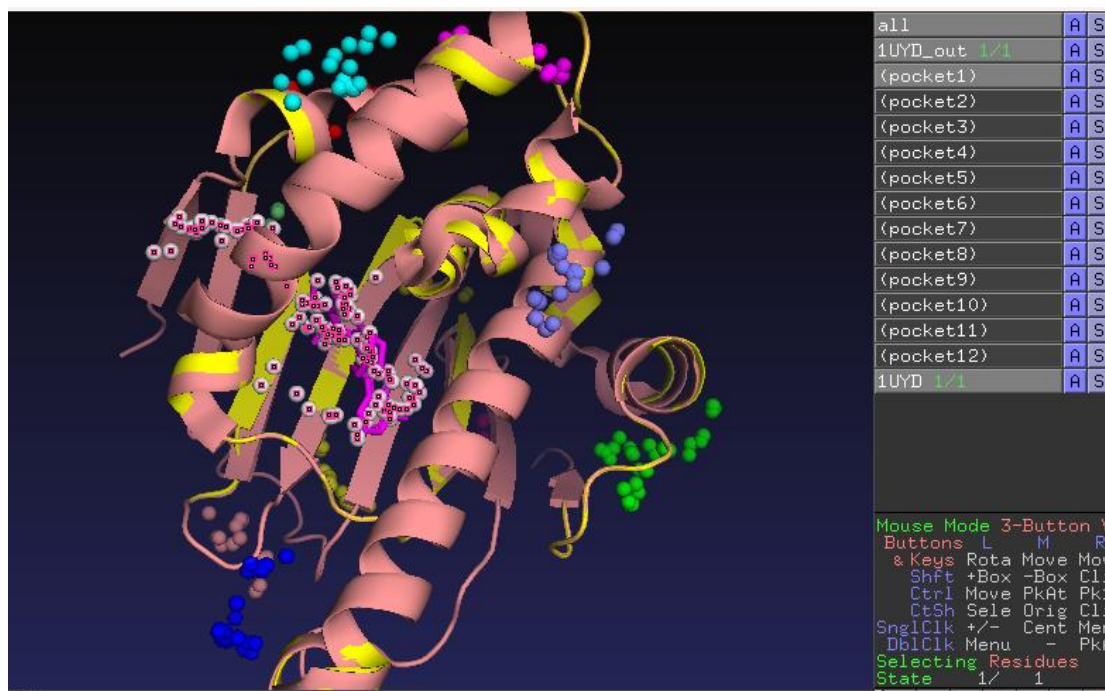


Figure 2

References

1. Krivák R, Hoksza D. P2Rank: machine learning based tool for rapid and accurate prediction of ligand binding sites from protein structure. J Cheminform. 2018;10(1):39. doi: 10.1186/s13321-018-0285-8.
2. Le Guilloux V, Schmidtke P, Tuffery P. Fpocket: An open source platform for ligand pocket detection. BMC Bioinformatics. 2009;10. doi: Artn 168 10.1186/1471-2105-10-168. PubMed PMID: WOS:000267596600001.